[TROUBLESHOOTING INTELLIGENT COMPACTION RESULTS]

As described previously, several jobsite conditions and operational factors can affect the results of intelligent compaction systems. With experience, operators will begin to recognize certain patterns and understand the probable cause of a deviation from expected values. Following are some common problems and related causes and solutions. Understanding this information will help you solve jobsite problems in a shorter period of time.

Problem: Compaction Measurement Values Are Lower Than Expected

Cause: Granular soil is too dry to compact. Added compaction results in soil structure are breaking down and de-compacting the soil.***Solution:*** Add moisture to the soil before compacting further. Granular soils can take a lot of water without becoming too wet, as the water tends to drain away. Add a bit more than ideally required to allow for dryout and drainage.

Cause: Soil composition is clay rather than gravel or granular material. Or, the clay can be buried below surface level and still affect the measurements.
Solution: Remove clay soil if practical, or accept lower values. Alternatively, utilize an energy-based, compactor-integrated compaction measurement technology such as MDP, which is not as affected by cohesive soils.

Cause: Drum is de-coupling on the hardest areas of ground. When the drum de-couples, RMVs are high and compaction measurement values (CMV) tend to read lower than the ground conditions would suggest.
Solution: Reduce amplitude to low setting. If still de-coupling, compaction is complete. Further compaction while de-coupling can cause de-compaction.

Cause: Clay soil is too wet.
Solution: Use a disk, harrow, or rotary mixer to dig up the soil so it can dry before attempting to compact it. Alternatively, utilize an energy-based compactor-integrated compaction measurement technology such as MDP which is not as affected by cohesive soils, but note that the moisture levels may still be unsuitable to achieve proper compaction.

Cause: Material being compacted was placed over a non-compacted or non-stabilized soil base. As a result it is flexing too much during compaction and will not compact.
Solution: The upper layer of soil will need to be removed and the lower layer of soil will need to be remedied. This might involve drying it out and re-compacting, adding lime or some other soil stabilizing agent, or even excavating the poor soil and replacing it.

Cause: The drum frequency is higher than it should be (this is unlikely).
Solution: Drum frequency should be near 30 Hz (1800 VPM) for the most consistent results. Have a mechanic determine why vibration speed is not performing properly and correct it. Alternatively, an energy-based, compactor-integrated compaction measurement technology such as MDP can be utilized in static mode (vibration off) to see if there is any effect on uniformity of compaction results.

Cause: There is a buried object or objects that are not as stiff as the surrounding soil, a pit where trees or other biomass are buried, buried refuse, or a clay ball. It will show up on the map as a relatively localized area.
Solution: Excavate and replace materials with good soil if the situation is serious enough to warrant it.

Cause: Travel speed is too fast.
Solution: Slow down to obtain most efficient productivity and higher compaction values. Utilize automatic speed control if the compactor is equipped with this option.

Cause: Travel direction affects the integrated compaction measurement values.
Solution: This is normal; the values will measure differently in forward than in reverse. There is no solution other than to drive only in one direction or accept the compaction values from only one direction during analysis.

Problem: Compaction Measurement Values Are Higher Than Expected

Cause: Base or sub-base is stiffer soil type than expected.
Solution: None, test using a dynamic cone penetrometer and check sub-soil shear strength. If stronger, accept results as normal.

Cause: Hidden object is buried below the surface. This might be a rock, concrete slab, old pavement, or building foundation.
Solution: Excavate object to achieve uniform compaction.

Cause: Travel speed is occasionally too slow. (This is unlikely unless the operator is trying to compact using a given number of passes).
Solution: Keep speed consistent. Utilize automatic speed control if the compactor is equipped with this option.

Cause: Ground is frozen.
Solution: None.

Problem: Compaction Measurement Values Are Erratic

Cause: Actual soil conditions are varying either at the surface or below surface level. This is more common than people assume. Hidden objects, changes in fill materials and varying water content can all affect compactor-integrated compaction measurement values.
Solution: If serious variations exist and need to be rectified, start with the easiest solutions first. Check soil moisture content and adjust. Excavate hidden objects if necessary and replace soil if critical.

Cause: Compactor-integrated compaction measurement values measured in forward are higher/lower than those obtained in reverse.
Solution: This is normal, and varies with soil type and compaction level. Typically the differences become smaller as the soil becomes more fully compacted.



Cause: Drum is de-coupling while compacting. De-coupling can cause wide variation in the compactor-integrated compaction measurement values, as the average values tend to drop when the drum begins de-coupling on harder ground.
Solution: Change to low amplitude. If de-coupling in low amplitude, the soil has reached the maximum stiffness the compactor can provide. Alternatively, an energy-based integrated roller compaction measurement technology such as MDP can be utilized in static mode (vibration off) to see if there is any effect on uniformity of compaction results.

[THE FUTURE OF INTELLIGENT COMPACTION]

As has been explained, measuring soil stiffness is extremely complex due to all of the variables involved. However, the more IC is used, the deeper your understanding of the capabilities and shortcomings of the technology. As your experience with it grows, new technologies, like Machine Drive Power, will emerge offering further benefits and addressing application issues that the last wave of progress has provided. As time passes, hardware solutions for existing technologies become less expensive, making them more accessible to repurpose for soil compaction applications.

The compactor of the future will likely have multiple measurement technologies available, as each technology has different useful capabilities. New measurement technologies will emerge—perhaps ground penetrating radar, ultrasound or magnetic imaging. Three-dimensional images documenting the entire road structure could be built. Moisture sensing technologies could alert an operator to call for a water truck or a ripper. Every compactor operator would have access to information from all machines on the jobsite (machine-to-machine communications). This provides real-time jobsite progress with advantages most easily seen with a multiple number of compactors or measuring devices. Jobsite superintendents can monitor and use the data to make the most cost-effective decisions on a daily basis.

Looking to the future, data will become more and more important. Data from yet-to-be-developed sensors, and the ability to quickly and easily take the data off the jobsite and into other off-board applications (PC, hand-held tablets, and others) will be an area of specific development. Current technologies already provide more data than can be easily sorted in a way that is useful to today's jobsite managers and inspectors. The capability of off-board data filtering, sorting and ability to create a jobsite report that meets the needs of the end-user will be of huge importance and an area where many specifications will soon be written.

It is an exciting era in the science of compaction and time will tell what advances and technologies will come next, but one thing is certain: With the cost savings, quality and efficiencies provided by intelligent compaction, the technologies will increasingly be required and included in specifications for jobsites worldwide.

APPENDIX

[SOIL CLASSIFICATION SYSTEMS]

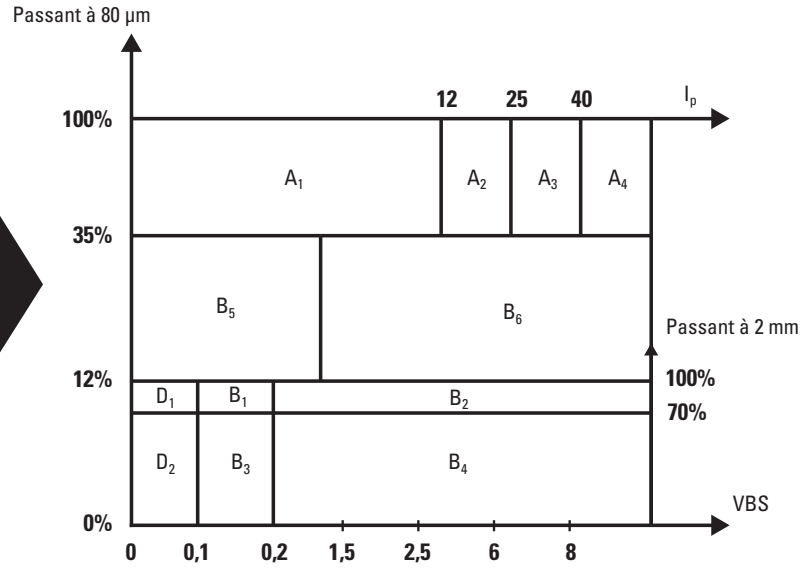
AASHTO CLASSIFICATION OF HIGHWAY SUBGRADE MATERIALS
(with suggested subgroups)

General Classification	Granular Materials (35% or less passing #200)							Silt-Clay Materials (more than 35% passing #200)			
Group Classification	A-1		A-3	A-2				A-4	A-5	A-6	A-7 A-7-5 A-7-6
	A-1-a	A-1-b		A-2-4	A-2-5	A-2-6	A-2-7				
Sieve Analysis Percent Passing:											
# 10	0-50										
#40	0-30	0-50	51-100								
#200	0-15	0-25	0-10	0-35	0-35	0-35	0-35	36-100	36-100	36-100	36-100
Characteristics of Fraction Passing #40:											
Liquid Limit				0-40	41+	0-40	41+	0-40	41+	0-40	41+
Plasticity Index	0-6		N.P.	0-10	0-10	11+	11+	0-10	0-10	11+	11+
Group Index	0		0	0		0-4		0-8	0-12	0-16	0-20
Usual Types of Significant Constituent Materials	Stone Fragments Gravel and Sand		Fine Sand	Silty or Clayey Gravel and Sand				Silty Soils		Clayey Soils	
General Rating as Subgrade	Excellent to Good					Fair to Poor					

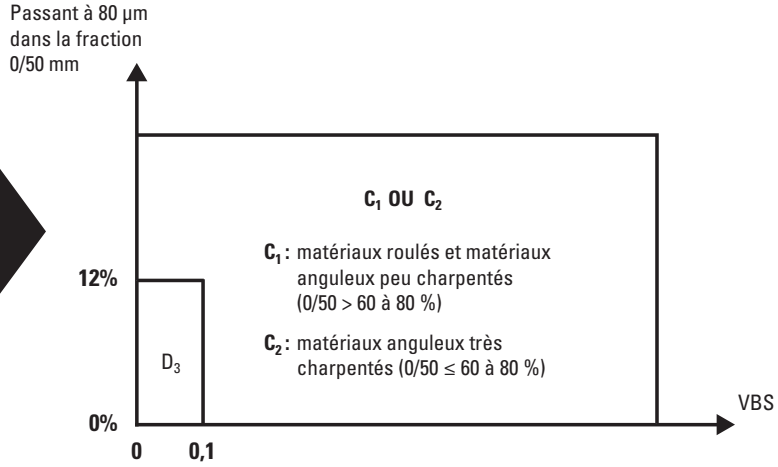
AASHTO Soil Classification System

Tableau synoptique de classification des matériaux selon leur nature

Sols
Dmax ≤ 50 mm



Sols
Dmax > 50 mm



Matériaux rocheux

Roches sédimentaires	Roches carbonatées	Craies	R_1
		Calcaires	R_2
	Roches argileuses	Marnes, argilites, pélites...	R_3
	Roches siliceuses	Grès, poudingues, brèches...	R_4
	Roches salines	Sel gemme, gypse	R_5
Roches magmatiques et métamorphiques	Granites, basaltes, andésites, gneiss, schistes métamorphiques et ardoisiers...		R_6

Matériaux particuliers

Sols organiques et sous-produits industriels	F

French Soil Classification System

Hauptgruppe	Korngrößenanteil ≤ 0,06 mm	Korngrößenanteil > 2,0 mm	Gruppe (allgemein)	Gruppe (detailliert)	Kurzzeichen Gruppensymbol
Grobkörniger Boden	≤ 5	< 40	Kies	Enggestufte Kiese	GE
				Weitgestufte Kies-Sand-Gemische	GW
				Intermittierend gestufte Kies-Sand-Gemische	GI
			Sand	Enggestufte Sande	SE
				Weitgestufte Sand-Kies-Gemische	SW
				Intermittierend gestufte Sand-Kies-Gemische	SI
Gemischtkörniger Boden	5 bis 40	< 40	Kies-Schluff	5 bis 15 Gew.-% ≤ 0,06 mm	GU
				15 bis 40 Gew.-% ≤ 0,06 mm	GU*
			Kies-Ton	5 bis 15 Gew.-% ≤ 0,06 mm	GT
				15 bis 40 Gew.-% ≤ 0,06 mm	GT*
		≤ 40	Sand-Schluff	5 bis 15 Gew.-% ≤ 0,06 mm	SU
				15 bis 40 Gew.-% ≤ 0,06 mm	SU*
			Sand-Ton	5 bis 15 Gew.-% ≤ 0,06 mm	ST
				15 bis 40 Gew.-% ≤ 0,06 mm	ST*
Feinkörniger Boden	< 40	—	Schluff	Leicht plastische Schluffe W_L≤35	UL
				Mittelplastische Schluffe W_L=35 bis 50	UM
			Ton	Leicht plastische Tone W_L≤35	TL
				Mittelplastische Tone W_L=35 bis 50	TM
				Ausgeprägt plastische Tone W_L=50	TA
Organogener Boden	< 40	—	Nicht brenn- und schwelbar	Organogene Schluffe W_L=35 bis 50	OU
				Organogene Tone W_L>50	OT
	≤ 40			Grob bis gemischtkörnige Böden mit humosen Beimengungen	OH
				Grob bis gemischtkörnige Böden mit kalkigen, kieseligen Bildungen	OK
Organischer Boden	—	—	Brenn- und schwelbar	Nicht bis mäßig zersetzte Torfe	HN
				Zersetzte Torfe	HZ
				Mudden (Faulschlamm)	F
Auffüllung[1]	—	—	—	Auffüllung aus Fremdstoffen	A

1 - Eine Auffüllung ist eine unter menschlicher Einwirkung entstandene Schüttung aus natürlichen Böden oder Fremdstoffen.

German Soil Classification System

USCS SOIL CLASSIFICATION SYSTEM

SOIL FRACTION	SYMBOL	SIZE RANGE
Boulders	**None**	Greater than 12"
Cobbles	**None**	75 mm (3") to 12"
1- Course Grained Soils:		
Gravel	**G**	75 mm (3") to #4 Sieve (4.25 mm)
Course Gravel		75 mm to 19 mm
Fine Gravel		#4 Sieve to 19 mm
Sand Course Sand Medium Sand Fine Sand	**S**	#4 Sieve to #200 Sieve (0.075 mm)
2- Fine Grained Soils:		
Fines		Less than #200 Sieve
Silt	**M**	Use Atterberg Limits
Clay	**C**	Use Atterberg Limits
3- Organic Soils	**O**	**Use Atterberg Limits**
4- Peat	**Pt**	**Visual Identification**
Gradation Symbols		**Liquid Limit Symbols**
Well-graded	**W**	High LL **H**
Poorly-graded	**P**	Low LL **L**

United Soil Classification System

SOIL GROUPS GRAVEL and SAND may be qualified sandy GRAVEL and gravely SAND where appropriate			SUB-GROUPS and in-laboratory identification			
			GROUP SYMBOL	SUB-GROUP SYMBOL	FINES % < 0.06 mm	LIQUID LIMIT
COARSE SOILS less than 35% of the material is finer than 0.06 mm	GRAVELS More than 50% of coarse material is of gravel size (coarser than 2 mm)	Slightly silty or clayey GRAVEL	G GW GP	GW GPu CPg	0 to 5	
		Silty GRAVEL Clayey GRAVEL	G-F G-M G-C	GWM GPM GWC GPC	5 to 15	
		Very silty GRAVEL Very clayey GRAVEL	GF GM GC	GML, etc. GCL GCI GCH GCV GCE	15 to 35	
	SANDS More than 50% of coarse material is of sand size (finer than 2 mm)	Slightly silty or clayey SAND	S SW SP	SW SPu SPg	0 to 5	
		Silty SAND Clayey SAND	S-F S-M S-C	SWM SPM SWC SPC	15 to 35	
		Very silty SAND Very clayey SAND	SF SM SC	SML, etc. SCL SCI SCH SCV SCE	15 to 35	
FINE SOILS more than 35% of the material is finer than 0.06 mm	Gravelly or sandy SILTS and CLAYS 35% to 65% fines	Gravelly SILT Gravelly CLAY	FG MG CG	MLG, etc. CLG CIG CHG CVG CEG		< 35 35 to 70 50 to 70 70 to 90 > 90
		Sandy SILT Sandy CLAY	FS MS CS	MLS, etc. CLS, etc.		
	SILTS and CLAYS 65% to 100% fines	SILT (M SOIL) CLAY	F M C	ML, etc. CL CI CH CV CE		< 35 35 to 70 50 to 70 70 to 90 > 90
ORGANIC SOILS		Description letter 'O' suffixed to say group or sub-group symbol		Organic matter in significant amount e.g. MHO – organic silt of high LL		
PEAT		Pt – consists predominantly of plant remains (fibrous or amorphous)				

Classification v1.00 Sept 2010

Primary Letter
G = Gravel
S = Sand
M = Silt
C = Clay
O = Organic Soil
Pt = Peat

Secondary Letter
W = Well graded
P = Poorly graded
M = With non-plastic fines
C = With plastic fines
L = Of low plasticity (LL < 50)
H = Of high plasticity (LL > 50)

United Kingdom Soil Classification System

[COMPACTION REQUIREMENTS]

Compactor integrated devices performing continuous compaction control soil application

BASIC/MINIMUM REQUIREMENTS					
One of the lower 3 blocks (one value)	At least 2 blocks	3 top blocks	Top block or 4 lower blocks	2 lower blocks	At least one block
		Time stamp	Close-loop mode		
Qualitative observation (ex: double-jump...)		Mapping on board	Actual N passes	Data post-treatment facilities, & additional information	
Dimensional bearing capacity (ex: modulus...)	Number of passes (actual vs. target value)	Automatic positioning on board 2D or 3D	Actual frequency	Result by histogram and statistics	Data exchange between machines
Dimensional (ex: stiffness...)	Relative evolution (% related to target values D or ND)	Manual positioning 2D + layer (optional)	Actual amplitude AO	Result by distance or surface	Remote data exchange
Non-dimensional value	End of compaction (D or ND)	Distance 1D	Actual speed V	Identification of machine and CCC device	Data exchange from office (USB stick)
1 Behaviour of the material (dynamic response)	**2** Status of compaction (Comparison)	**3** Positioning, traceability during process	**4** Operational information (record and display)	**5** Control report, documentation	**6** Communication, others

From the pamplet *CECE – Guidelines to evaluate soil and asphalt compactors equipped with continuous compaction control (CCC)*

GLOSSARY OF TERMS

- A -

Aggregate	The granular, load-bearing mineral component of a road structure, usually sand, gravel, shells, slag, crushed stone or fines.
Amplitude	The measure of one-half of the total peak-to-peak vertical movement of a vibrating drum per complete cycle.
Atterberg Limits Method	A set of standards that describe seven stages of soil characteristics moving from a solid to a liquid. The most important stages are the Plastic Limit and the Liquid Limit.

- B -

Base	Also referred to as the "base course"; a layer of specified or selected material of planned thickness constructed on the sub-base for the purpose of serving one or more functions including distributing the load, providing drainage and minimizing frost action.

- C -

Calibration	The process of adjusting the parameters of the system to maximize its capabilities for use in the site material.
Capillarity	The ability of a material to absorb water upward or laterally.
Centrifugal Force	The force that causes a rotating, unbalanced weight to accelerate away from its axis.
Clay	A fine-grained mineral material (soil) that uses electro-chemical surface charges to bond well with water.
Coarse-grained Soil	A classification for soil comprised of particles (grains) that lack cohesion. Sand and gravel are considered coarse-grained soils. Coarse-grained soils are defined as well-graded or poorly-graded, which reflect the ability of the soils to be compacted.
Cohesion	The ability of a material to maintain its strength when unconfined; i.e. cling together and maintain its form through changes in moisture content or submersion.
Compaction	The process of reducing voids in a material through the use of mechanical manipulation; increasing the density.
Compactability	The degree to which a soil material is able to overcome material-specific deformation resistance.
Compaction Meter Value (CMV)	An indication of soil stiffness calculated by measuring the G-forces at the vibratory frequency of the drum and at the first harmonic (2x the vibratory frequency of the drum).
Compaction Meter Value Method	A method of indicating soil stiffness invented by the Swiss firm Geodynamik and utilized by Caterpillar.
Compressibility	The rate at which a material reduces in volume when a force is applied to it.

- D -

De-compaction	A loss of compaction due to re-application of unnecessary compactive force.
De-coupling	Also referred to as "double-jumping"; is a phenomenon where the drum rebounds from a vibratory impact and bounces high enough to allow the next vibration to occur while the drum is still in the air.
Density	The measure of mass per unit of volume; a traditional indicator of load-bearing strength.
Double-jumping	See De-coupling.

- E -

Elasticity	The tendency of a material to return to its original (or near original) shape after a compressive load is removed.
Embankment	Any fill whose top is higher than the adjoining surface.
Energy Method	The principle behind Machine Drive Power (MDP), a Caterpillar proprietary method of determining compaction by measuring the energy required to drive over the soil (rolling resistance).

- F -

Fines	Generally, materials of very small particle size falling below a certain threshold established by sieve testing. Fines will pass through the smallest sieve. Organizations around the world have independent definitions of the precise sieve size, but they are approximately the same.
Fine-grained Soils	Soils composed predominantly of fines.
Force/Displacement Method	A method of indicating soil stiffness by using drum characteristics and drum acceleration measurements to calculate drum displacement.
Frequency	A measure of the number of complete cycles (e.g. vibrations) over a given length of time.

- G -

Global Navigation Satellite System (GNSS)	An overarching term used to describe satellite based mapping technologies, including GPS and GLONASS.
Global Positioning System (GPS)	A US controlled worldwide radio-navigation system composed of a constellation of 24 satellites and their ground stations.
GLONASS	A Russian satellite constellation similar to GPS.
Grade	The inclination of a surface.
Gradation	The size range of individual soil particles.
Gradient of Compaction	The degree of compaction throughout the depth of influence. Soil will tend to be less compact at the surface, be more compact through a large central zone, and become less compact again toward the extreme depth of influence.
Grain	A mineral particle.
Grain-size Distribution	The measure of the range and distribution of different particle sizes in a soil.
Gravel	A coarse-grained mineral material; defined by the USCS as those particles less than 75 mm (3 in) in diameter not passing through a #4 sieve.

- I -

Impact A force of increased magnitude created by causing static pressure to become dynamic, e.g. dropping a weight. Low-frequency or irregular frequency blows (blows of 50-600 blows per minute) are considered impact forces.

Intelligent Compaction Generally speaking, intelligent compaction can be defined as a compactor integrated technology applied to the compaction process that improves worksite efficiencies by eliminating human guesswork.

- L -

Lift A single layer of placed soil material. It can vary in thickness.

Linear Load The measure used by the industry to compare the compaction potential of static smooth drum compactors.

Liquid Limit A highly significant Atterberg Limit; the point at which a soil contains so much water that it is considered a liquid.

Load-bearing Strength The ability of a material to support a load.

- M -

Machine-integrated Compaction Measurement Conducting compaction measurement with technologies incorporated into the machine performing the compaction.

Manipulation A kneading process that rearranges particles into a more dense mass.

Moisture Content The amount of liquid (water) per volume of a mass.

- N -

Natural Frequency The frequency at which a mass vibrates due to its own characteristics.

- P -

Particle Distribution See Grain-size Distribution.

Pass The number of times a compactor will pass over an area of ground. Sometimes, a "pass" will be defined as a round trip forward and back, passing over a particular area twice; and other times a pass will mean a single instance that a compactor travels over an area. Caterpillar defines a pass as a single trip over an area in either a forward or reverse direction.

Permeability A material's ability to allow the passage of a gas or liquid.

Plasticity The property of a fine-grain soil that allows it to deform beyond the point of recovery without cracking or appreciable volume change.

Plasticity Index The difference between the Liquid Limit and the Plastic Limit of a soil. This measure is used to determine the extent of soil stabilization required for fine-grained soils.

Plastic Limit A highly significant Atterberg Limit; the point at which a soil retains enough moisture to become plastic.

Poorly-graded	Also referred to as uniformly-graded; the quality of coarse-grained soils to contain particles of relatively uniform size, making it difficult to compact.
Proctor Test (Standard or Modified)	A laboratory test that determines the maximum dry density of a soil material as well as the optimum water content to achieve maximum density.
Production Mode	A compactor system setting that optimizes system useability for high-production applications where accuracy is not the main goal.
Proofing Mode	A compactor system setting that optimizes system useability for high-accuracy applications where machine productivity is not the main goal.

- Q -

Quality Assurance (QA)	The testing methods and data that the project owner uses to document the quality of the compaction achieved on a project.
Quality Control (QC)	The procedure a contractor implements to ensure that the compaction work will be completed to specification.

- R -

Remediation	The process of altering the soil by chemical or mechanical means to improve its engineering properties.
Resonance	The convergence of the vibratory frequencies of two vibrating masses.
Resonance Meter Value (RMV)	An indication of the degree to which the drum is de-coupling.
Resonant Frequency	For vibratory compactors, the point at which vibrating soil material contributes to the vibration of the compactor enough to cause the compactive effort to exceed the centrifugal force generated; i.e. the output is greater than the effort.
Rolling Resistance	The amount of energy required to roll a round shape through a material.

- S -

Sand	A non-cohesive mineral particle of a defined size and shape.
Settlement	The process of decreasing surface elevation due to consolidation of fill material.
Shear Resistance	The ability of a soil's particles to resist sliding across each other when a compactive force is applied.
Silt	A non-cohesive, fine-grained mineral material (soil).
Soil	Unconsolidated material composed of mineral particles that may or may not contain organic substances.
Soil Stabilization	The process of maximizing the suitability of soil for a given construction purpose.
Static Pressure	A weight applied to the compaction effort.
Station	A non-standard area defined by engineers and marked by stakes at the jobsite for the purpose of controlling preparation by manageable sections.
Stiffness (Soil)	The ability of a material (soil) to resist deflection under load; a primary indicator of load-bearing strength.
Sub-base	A layer between the sub-grade and base.
Sub-grade	The soil prepared to support a traffic structure. Essentially, it performs as the foundation of the structure, and is sometimes referred to as "basement soil" or "foundation soil."

- T -

Texture A quality that defines the surface friction of a soil particle.

Total Applied Force A calculation of the maximum amount of vibratory energy that a compactor can apply to the ground.

- U -

Uniformity The maintenance of consistency in materials and applications.

Uniformity Coefficient A parameter to describe the particle size distribution (grading curve) of a soil.

Uniformly-graded Also referred to as poorly-graded; the quality of coarse-grained soils to contain particles of relatively uniform size, making them difficult to compact.

- V -

Vibration A high-frequency series of blows (1400-4000 blows per minute) that produce a rapid succession of pressure waves. Vibrations produced by a compactor can break the bonds between particles of a material that is being compacted.

Void The space in a volume of material not occupied by solid mineral material.

- W -

Well-graded The quality of coarse-grained soils to contain particles of many sizes, making them easier to compact.




CAT CS78B


CATERPILLAR